📊 Detailed Question-Level Report

Select Question

28840431

Question ID: 28840431

Difficulty: Easy

A cyclist maintains a constant speed of 25 km/h. How much distance will they cover in 3.5 hours?

Options

A. 87.5 km

B. 85 km

C. 90 km

D. 82.5 km

Correct Answer

A

Canonical Solution

87.5 km

Solver A Output

```
{
  "equation_model" : "Distance = Speed × Time"
  "derivation" : "Given:
                 Speed (S) = 25 km/h
                 Time (T) = 3.5 hours

                 Using the formula for distance:
                 Distance (D) = S × T
                 D = 25 km/h × 3.5 hours
                 D = 87.5 km"
  "answer" : "87.5 km"
  "confidence" : 1
}
```

Solver B Output

```
{
  "method" : "Plug-in verification (Reverse Reasoning)"
```

"steps" :
"The problem states a cyclist's constant speed is 25 km/h and the time traveled is 3.5 hours. We need to find the total distance covered. Instead of directly calculating Distance = Speed × Time, we will use reverse reasoning by testing each given distance option. For each option, we will calculate the implied speed using the formula Speed = Distance / Time, and check if it matches the given speed of 25 km/h.

1. Assume the first option (87.5 km) is the correct distance. Calculate the implied speed:
   Implied Speed = 87.5 km / 3.5 hours = 25 km/h.
   This matches the given speed of 25 km/h. Thus, 87.5 km is a consistent answer.

2. To confirm, let's briefly check other options:
   - If the distance were 85 km: Implied Speed = 85 km / 3.5 hours ≈ 24.29 km/h. This is not 25 km/h.
   - If the distance were 90 km: Implied Speed = 90 km / 3.5 hours ≈ 25.71 km/h. This is not 25 km/h.
   - If the distance were 82.5 km: Implied Speed = 82.5 km / 3.5 hours ≈ 23.57 km/h. This is not 25 km/h.

Only 87.5 km produces the correct speed when divided by the given time."
"answer" : "87.5 km"
"confidence" : 1
}

🔍 Validation Details

{
"match_A" : true
"match_B" : true
"final_validation" : "valid"
"reasoning" :
"Both Solver A and Solver B provided an answer of '87.5 km', which is identical to the generator's answer in both numeric value and unit."
}

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